

Network Planning

- Understanding the topology
- Device roles (ISP, Edge Router, ASA Firewall, Layer 3 Switch, Layer 2 Switches, DHCP Server, Wireless Router)
- IP addressing plan
- VLAN design
- Cable selection

Physical Connections

- Exactly which interface on each device connects to which interface on the next device (e.g., ISR4331 Gi0/0/0 → ASA Gi1/1, ASA Gi1/2 → 3560 Gi0/1, etc.)

Layer 3 Switch Configuration

- Create VLANs
- Configure VLAN names
- Configure SVIs
- Enable ip routing
- Configure trunk links
- Configure access ports
- Verify VLANs and trunks

Layer 2 Switch Configuration (each switch individually)

- Management Switch
- HR Switch
- Finance Switch
- IT Switch
- Assign access ports
- Configure trunk uplinks
- Verification

DHCP Server Configuration

- Assign static IP
- Configure multiple DHCP pools
- Configure gateways
- Configure DNS

- Configure excluded addresses
- Verify leases

DHCP Relay (ip helper-address)

- Why DHCP broadcasts do not cross VLANs
- Configure ip helper-address on every SVI
- Verify relay operation

Edge Router Configuration

- Interface IP addresses
- Default route
- Static routes (or OSPF as an enhancement)
- Verification

ASA Firewall Configuration

- Configure inside and outside interfaces
- Default route
- NAT
- ACLs
- Block EX1, EX2, EX3
- Permit internal traffic
- Verify logs and connectivity

ISP Router Configuration

- WAN addressing
- Routes to the enterprise network
- Internet simulation

Wireless Router Configuration

- Guest VLAN
- SSID
- WPA2 security
- DHCP disabled (since the dedicated server provides addresses)

End Device Configuration

- Set PCs to DHCP
- Verify IP addresses
- Verify gateways
- Verify DNS

Testing

- Ping within the same VLAN
- Ping between VLANs
- Ping the DHCP server
- Ping the firewall
- Ping the ISP router
- Verify EX1/EX2/EX3 are blocked
- Use Simulation Mode to observe DHCP and ICMP packets

Troubleshooting

- Common mistakes
- Useful show commands
- How to diagnose VLAN, trunk, routing, DHCP, and firewall issues